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United States Patent Application Publication

20250282574

Kind Code

A1

Publication Date

September 11, 2025

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Autonomous Artificial Muscle Fiber Coiler System and Method of Use

Abstract

The present invention is an autonomous artificial muscle fiber coiler system. The system comprises a motor-driven spool holder for feeding monofilament material under controlled tension, a pulley system to regulate tension dynamically, and a bevel gear assembly with synchronized large bevel gears and a smaller third bevel gear to twist the monofilament into a desired helical structure. Two 'mirrored' or parallel motor drivers, controlled by a power supply unit, vary the speed and direction of the gears to achieve customized twist density. The system is designed to produce consistent and precise coiled fibers for applications in robotics, prosthetics, and advanced materials. The coiler can also produce graded fibers by dynamically adjusting tension and twist density. The system supports multiple materials, including nylon, polyester, Kevlar, and conductive polymers.

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Family ID: 1000008508351

Appl. No.: 19/068088

Filed: March 03, 2025

Related U.S. Application Data

us-provisional-application US 63561766 20240306

Publication Classification

Int. Cl.: B65H54/00 (20060101)

U.S. Cl.:

CPC B65H54/00 (20130101); B65H2403/43 (20130101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] The present application claims priority to, and the benefit of, U.S. Provisional Application No. 63/561,766 which was filed on Mar. 6, 2024, and is incorporated herein by reference in its entirety.

FIELD OF THE INVENTION

[0002] The present invention generally relates to the field of autonomous systems for manufacturing artificial muscle fibers. More specifically, the present invention relates to an innovative artificial muscle fiber coiler system designed to automate the process of producing coiled fibers for applications in robotics, prosthetics, and advanced materials. The system includes a motor-driven spool holder for feeding monofilaments with controlled tension, a pulley system for dynamic tension adjustment, and a bevel gear assembly for twisting the monofilament into a helical structure. The system further features 'mirrored' or parallel motor drivers for precise control of twist density and a power supply unit for stable operation. Accordingly, the present disclosure makes specific reference thereto. Nonetheless, it is to be appreciated that aspects of the present invention are also equally applicable to other like applications, devices, and methods of manufacture.

BACKGROUND

[0003] By way of background, robotic muscle fibers are an integral component in the development of advanced robotics, prosthetics, and other fields that demand high-performance, lightweight actuators. The artificial fibers mimic the functionality of biological muscles, offering versatility in applications ranging from industrial automation to medical devices. However, the traditional methods of creating artificial muscle fibers are labor-intensive and prone to inconsistencies.

[0004] Conventionally, the robotic muscle fibers have been manually created using power tools. The conventional process is not only time-consuming but also requires significant manual dexterity and precision. The laborious nature of the work often results in limited fiber length, as longer fibers are more susceptible to snapping during coiling. Irregularities in the coiling process, such as uneven tension and irregular twists, frequently lead to fiber breakage, requiring additional time and effort to repair or restart the process.

[0005] Moreover, manual techniques do not provide the level of precision needed to create fibers with uniform or graded mechanical properties, which are often required for specific applications. These limitations have hindered the scalability and reliability of artificial muscle fiber production, creating a demand for an autonomous solution that eliminates manual labor, improves consistency, and minimizes material waste.

[0006] Therefore, there exists a long-felt need in the art for an autonomous system that can efficiently and consistently produce artificial muscle fibers. There is also a long-felt need in the art for a system that eliminates the labor-intensive and error-prone process of manually coiling artificial muscle fibers. Additionally, there is a long-felt need in the art for a system that provides uniformity and precision in coiled fibers. Moreover, there is a long-felt need in the art for a solution that can dynamically control tension and twist density to produce artificial muscle fibers. Further, there is a long-felt need in the art for a system that can handle a wide range of monofilament materials, including nylon, polyester, Kevlar, and conductive polymers, to support diverse applications. Furthermore, there is a long-felt need in the art for a robust, scalable system capable of producing artificial muscle fibers in continuous lengths without frequent interruptions for adjustments or repairs. Finally, there is a long-felt need in the art for an autonomous artificial muscle fiber coiler system that enhances efficiency, precision, and versatility in fiber production while minimizing material waste and manual intervention.

[0007] The subject matter disclosed and claimed herein, in one embodiment thereof, comprises an autonomous artificial muscle fiber coiler system designed to address the aforementioned challenges. The coiler system comprises a motor-driven spool holder configured to hold a monofilament material and feed it steadily and under controlled tension into subsequent stages of the system. The system further includes a pulley system with a plurality of pulleys to provide dynamic tension control and regulate the length of the monofilament where coiling occurs. A bevel gear assembly, consisting of two large bevel gears and a smaller third bevel gear positioned between them, twists the monofilament into a helical structure. Two 'mirrored' or parallel motor drivers drive the bevel gear assembly that dynamically adjust the speed and direction of the gears to control the coiling process. A power supply unit provides electrical power to the system and includes control knobs for fine adjustments of voltage and supplied current to ensure optimal operation.

[0008] In this manner, the autonomous artificial muscle fiber coiler system of the present invention accomplishes all of the foregoing objectives and provides users with an innovative solution for producing artificial muscle fibers. The system automates the entire process of coiling monofilaments, significantly reducing the time and effort required compared to manual methods. The motor-driven spool holder ensures steady feeding and prevents slack or irregularities in the fiber, while the pulley system dynamically adjusts tension to produce consistent and reliable coils. The bevel gear assembly precisely twists the monofilament, and its adjustable speed and direction enable the production of fibers with uniform or graded mechanical properties. The system is compatible with a variety of monofilament materials, including nylon, polyester, Kevlar, and conductive polymers, making it highly versatile.

SUMMARY OF THE INVENTION

[0009] The following presents a simplified summary in order to provide a basic understanding of some aspects of the disclosed innovation. This summary is not an extensive overview, and it is not intended to identify key/critical elements or to delineate the scope thereof. Its sole purpose is to present some general concepts in a simplified form as a prelude to the more detailed description that is presented later.

[0010] The subject matter disclosed and claimed herein, in one embodiment thereof, comprises an autonomous system for coiling artificial muscle fibers. The system comprises a material feeder configured to hold and dispense a monofilament, a pulley assembly is configured to regulate the tension of the monofilament during processing, a bevel gear assembly driven by motors to impart a coiled structure to the monofilament, and at least one motor driver is configured to operate the gear assembly in synchronization, wherein the system produces a continuous coiled artificial muscle fiber with minimized irregularities and breakage.

[0011] In yet another embodiment, an artificial muscle fiber coiler system is disclosed. The system includes a motor-driven spool holder adapted to hold a monofilament material and feed the monofilament steadily and under controlled tension, a pulley system comprises a plurality of pulleys configured to provide dynamic tension and regulate the length of the monofilament where coiling occurs, a bevel gear assembly comprises a first large bevel gear and a second large bevel gear rotating synchronously, and a smaller third bevel gear disposed between the first and second large bevel gears, is configured to synchronize the rotation of the large bevel gears and control the twisting process of the monofilament. Two 'mirrored' or parallel motor drivers are configured to drive the bevel gear assembly and vary the speed and direction of the smaller third bevel gear to twist the monofilament into a desired coiled structure. A power supply unit is operably coupled to the motor-driven spool holder and the motor drivers, configured to provide electrical power for the system's operation.

[0012] In another embodiment, a method for coiling artificial muscle fibers using an artificial muscle fiber coiler system is disclosed. The method comprises the steps of rotating a spool holder to unwind a monofilament material, transferring the monofilament to a pulley system to provide

consistent tension and regulate the length of the monofilament for coiling, driving a bevel gear assembly comprising two large bevel gears and a smaller third bevel gear disposed between them to twist the monofilament into a helical structure by synchronizing the rotation of the large bevel gears, and winding the twisted monofilament onto a take-up spool to form a continuous coiled fiber.

[0013] In still another embodiment, the motor-driven spool holder adjusts the tension of the monofilament in real-time to prevent slack and ensure uniform feeding into the pulley system.

[0014] In another aspect, the pulley system is configured to dynamically adjust tension based on pre-set tension profiles to create artificial muscle fibers with graded mechanical properties along their length.

[0015] In yet another aspect, the bevel gear assembly is made of a material selected from the group consisting of steel, brass, aluminum, and plastic, and the smaller third bevel gear is configured with a gear ratio of up to 2:1 relative to the large bevel gears for enhanced torque during the twisting process.

[0016] Numerous benefits and advantages of this invention will become apparent to those skilled in the art to which it pertains upon reading and understanding of the following detailed specification.

[0017] To the accomplishment of the foregoing and related ends, certain illustrative aspects of the disclosed innovation are described herein in connection with the following description and the annexed drawings. These aspects are indicative, however, of but a few of the various ways in which the principles disclosed herein can be employed and are intended to include all such aspects and their equivalents. Other advantages and novel features will become apparent from the following detailed description when considered in conjunction with the drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0018] The description refers to provided drawings in which similar reference characters refer to similar parts throughout the different views, and in which:

[0019] FIG. 1 illustrates a perspective view of the artificial muscle fiber coiler system of the present invention in accordance with the disclosed structure;

[0020] FIG. 2 illustrates a perspective view of the bevel gear assembly used in the artificial muscle fiber coiler system of the present invention in accordance with the disclosed structure; and

[0021] FIG. 3 illustrates a flow chart depicting a process of coiling monofilament performed by the artificial muscle fiber coiler system of the present invention.

DETAILED DESCRIPTION OF THE PRESENT INVENTION

[0022] The innovation is now described with reference to the drawings, wherein like reference numerals are used to refer to like elements throughout. In the following description, for purposes of explanation, numerous specific details are set forth in order to provide a thorough understanding thereof. It may be evident, however, that the innovation can be practiced without these specific details. In other instances, well-known structures and devices are shown in block diagram form in order to facilitate a description thereof. Various embodiments are discussed hereinafter. It should be noted that the figures are described only to facilitate the description of the embodiments. They are not intended as an exhaustive description of the invention and do not limit the scope of the invention. Additionally, an illustrated embodiment need not have all the aspects or advantages shown. Thus, in other embodiments, any of the features described herein from different embodiments may be combined.

[0023] As noted above, there exists a long-felt need in the art for an autonomous system that can efficiently and consistently produce artificial muscle fibers. There is also a long-felt need in the art for a system that eliminates the labor-intensive and error-prone process of manually coiling

artificial muscle fibers. Additionally, there is a long-felt need in the art for a system that provides uniformity and precision in coiled fibers. Moreover, there is a long-felt need in the art for a solution that can dynamically control tension and twist density to produce artificial muscle fibers. Further, there is a long-felt need in the art for a system that can handle a wide range of monofilament materials, including nylon, polyester, Kevlar, and conductive polymers, to support diverse applications. Furthermore, there is a long-felt need in the art for a robust, scalable system capable of producing artificial muscle fibers in continuous lengths without frequent interruptions for adjustments or repairs. Finally, there is a long-felt need in the art for an autonomous artificial muscle fiber coiler system that enhances efficiency, precision, and versatility in fiber production while minimizing material waste and manual intervention.

[0024] The present invention, in one exemplary embodiment, is an artificial muscle fiber coiler system. The system includes a motor-driven spool holder adapted to hold a monofilament material and feed the monofilament steadily and under controlled tension, a pulley system comprises a plurality of pulleys configured to provide dynamic tension and regulate the length of the monofilament where coiling occurs, a bevel gear assembly comprises a first large bevel gear and a second large bevel gear rotating synchronously, and a smaller third bevel gear disposed between the first and second large bevel gears, is configured to synchronize the rotation of the large bevel gears and control the twisting process of the monofilament. Two ‘mirrored’ or parallel motor drivers are configured to drive the bevel gear assembly and vary the speed and direction of the smaller third bevel gear to twist the monofilament into a desired coiled structure.

[0025] Reference will now be made in detail to the present preferred embodiments of the invention, examples of which are illustrated in the accompanying drawings. Wherever possible, the same reference numerals are used in the drawings and the description to refer to the same or like parts.

[0026] Referring initially to the drawings, FIG. 1 illustrates a perspective view of the artificial muscle fiber coiler system of the present invention in accordance with the disclosed structure. The autonomous artificial muscle fiber coiler system **100** of the present invention is designed to automate the process of coiling artificial muscle fibers for various applications like robotics and prosthetics. More specifically, the system **100** employs a robotic or autonomous system to perform the coiling process, eliminating the need for manual labor. Further, the system **100** provides consistent and precise coiling, minimizing irregularities and fiber breakage. The system **100** includes a motor-driven spool holder **102**. The motor-driven spool holder **102** is adapted to hold the raw monofilament material/fiber **104** for the coiling process and feeds the monofilament steadily and under controlled tension into the next stages of the system **100**. The motor-driven spool holder **102** also minimizes slack and prevents irregularities in the fiber. Monofilament including but not limited to nylon, polyester, Kevlar, conductive polymers, elastomers, carbon fiber filaments, and more can be used in the system **100**.

[0027] A power supply unit **106** provides electric power to different electronic components of the system **100**. The unit **106** includes a plurality of controllers **108** for fine control over the supplied voltage and supplied current. The power supply unit **106** may include batteries or can also be connected to conventional AC power supply for providing electrical supply to the artificial muscle fiber manufacturing system **100**. The power supply unit **106** preferably provides direct current compatibility with different motors used in the system **100**.

[0028] The monofilament **104** is fed to a pulley system **110** of the artificial muscle fiber coiler system **100**. The pulley system **110** includes a plurality of pulleys **112** for providing dynamic tensioning and movement for the monofilament. The pulleys **112** rotate synchronously for regulating the fiber's tension and controlling the length of the fiber where the coiling occurs. The pulley system **110** also eliminates issues with uneven coils or fiber breaks, ensuring uniformity and reliability in the artificial muscle fibers.

[0029] A bevel gear assembly **114** is designed to twist the monofilament into the desired coiled structure. The bevel gear assembly **114** includes a first large bevel gear **115** and a second large

bevel gear **116**. The large bevel gears **115**, **116** rotate for twisting the monofilament and a smaller third bevel gear **118** is disposed between the two large bevel gears **115**, **116**. The smaller third bevel gear **118** is configured to synchronize rotation of the large bevel gears **115**, **116** and control the twisting process of the monofilament.

[0030] Two ‘mirrored’ or parallel motor drivers **120a**, **120b** are disposed in the artificial muscle fiber manufacturing system **100** and are used for driving the motors used for the bevel gear assembly **114**. The motor drivers **120** (i.e., **120a**, **120b**) are coupled to the power supply unit **106** and regulates the power supply for fine control over motor speed and direction. The motor drivers **120** (i.e., **120a**, **120b**) may include a plurality of electronic components such as resistors, capacitors, and more. The motor drivers **120** (i.e., **120a**, **120b**) vary the speed of driving of the smaller gear **118**, thereby twisting the monofilament fiber. Preferably, the system **100** is portable and can be placed on a platform **124** which can be positioned on any surface for easy placement and portability. The platform **124** can be made of glass, metal, wood, or any other durable and lightweight material.

[0031] FIG. 2 illustrates a perspective view of the bevel gear assembly used in the artificial muscle fiber coiler system of the present invention in accordance with the disclosed structure. The bevel gear assembly **114** is disposed on a frame **202** which includes a power connector **204** for connecting to at least one motor driver. The large bevel gears **115**, **116** preferably rotate synchronously and the smaller third bevel gear **118** twists the monofilament. The smaller third bevel gear **118** rotates non-synchronously with the large bevel gears **115**, **116**. Referring again to FIG. 1, the bevel gear assembly **114** includes a supporting member **122** for supporting the assembly **114** and preventing lateral movement of the bevel gears.

[0032] FIG. 3 illustrates a flow chart depicting a process of coiling monofilament performed by the autonomous artificial muscle fiber coiler system of the present invention. Initially, the spool holder **102** rotates and unwinds the monofilament therefrom (Step **302**). The monofilament is continuously transferred to the pulley system **110** where the plurality of pulleys **112** provides a consistent tension in the monofilament (Step **304**). Then, the bevel gear assembly **114** twists the monofilament into the desired helical structure (Step **306**), and the coiled filament is wound onto a take-up spool for storage (Step **308**).

[0033] In some embodiments, the pulley system **110** dynamically adjusts the tension of the monofilament based on a pre-set profile and the bevel gear system **114** twists the monofilament while maintaining the desired tension variation. The system **100** of the present invention can produce artificial muscle fibers with uniform mechanical properties or graded mechanical properties such as stronger at one end and more elastic at the other. For graded mechanical properties, the motor speeds in the bevel gear system are adjusted dynamically to vary the twist density of the monofilament.

[0034] In some embodiments of the present invention, two or more spools of monofilament are used in the autonomous artificial muscle fiber coiler system **100** wherein different sets of pulleys maintain synchronized tension for all materials and the bevel gear assembly **114** twists the materials together into a composite artificial muscle fiber.

[0035] The bevel gear assembly **114** can be made of steel, brass, aluminum, plastic, or any other similar material. The preferable gear ratio for the different gears of the assembly **114** is 1:1 and the smaller gear may have gear ratio from 1:1 to 2:1. The larger bevel gears **115**, **116** may have 30-40 teeth per inch and the smaller third bevel gear **118** may have 15-20 teeth per inch.

[0036] Certain terms are used throughout the following description and claims to refer to particular features or components. As one skilled in the art will appreciate, different persons may refer to the same feature or component by different names. This document does not intend to distinguish between components or features that differ in name but not structure or function. As used herein “autonomous artificial muscle fiber coiler system”, “artificial muscle fiber coiler system”, “artificial muscle fiber manufacturing system”, and “system” are interchangeable and refer to the

autonomous artificial muscle fiber coiler system **100** of the present invention.

[0037] Notwithstanding the forgoing, the autonomous artificial muscle fiber coiler system **100** of the present invention can be of any suitable size and configuration as is known in the art without affecting the overall concept of the invention, provided that it accomplishes the above stated objectives. One of ordinary skill in the art will appreciate that the autonomous artificial muscle fiber coiler system **100** as shown in the FIGS. are for illustrative purposes only, and that many other sizes and shapes of the autonomous artificial muscle fiber coiler system **100** are well within the scope of the present disclosure. Although the dimensions of the autonomous artificial muscle fiber coiler system **100** are important design parameters for user convenience, the autonomous artificial muscle fiber coiler system **100** may be of any size that ensures optimal performance during use and/or that suits the user's needs and/or preferences.

[0038] Various modifications and additions can be made to the exemplary embodiments discussed without departing from the scope of the present invention. While the embodiments described above refer to particular features, the scope of this invention also includes embodiments having different combinations of features and embodiments that do not include all of the described features. Accordingly, the scope of the present invention is intended to embrace all such alternatives, modifications, and variations as fall within the scope of the claims, together with all equivalents thereof.

[0039] What has been described above includes examples of the claimed subject matter. It is, of course, not possible to describe every conceivable combination of components or methodologies for purposes of describing the claimed subject matter, but one of ordinary skill in the art may recognize that many further combinations and permutations of the claimed subject matter are possible. Accordingly, the claimed subject matter is intended to embrace all such alterations, modifications and variations that fall within the spirit and scope of the appended claims.

Furthermore, to the extent that the term “includes” is used in either the detailed description or the claims, such term is intended to be inclusive in a manner similar to the term “comprising” as “comprising” is interpreted when employed as a transitional word in a claim.

Claims

1. An autonomous artificial muscle fiber coiler system comprising: a motor-driven spool holder; a monofilament fiber; a power supply unit; a plurality of controllers; a pulley system; and a bevel gear assembly; wherein said motor-driven spool holder holds said monofilament fiber; wherein said monofilament fiber is fed to said pulley system of the artificial muscle fiber coiler system; wherein said pulley system having a plurality of pulleys for providing dynamic tensioning and twisting movement of said monofilament fiber; wherein said plurality of pulleys rotate synchronously for regulating said monofilament fiber dynamic tension and for controlling a length of said monofilament fiber for forming an artificial muscle fiber; wherein said bevel gear assembly twists said monofilament fiber into a coiled structure; wherein said bevel gear assembly having a first bevel gear and a second bevel gear rotating for the twisting movement of said monofilament fiber; wherein said bevel gear assembly having a third bevel gear disposed between said first bevel gear and said second bevel gear; wherein said first bevel gear and said second bevel gear are larger than said third bevel gear; and further wherein said third bevel gear synchronizes rotation of said first bevel gear and said second bevel gear and controls the twists of said monofilament fiber.
2. The autonomous artificial muscle fiber coiler system of claim 1, wherein said bevel gear assembly having two parallel motor drivers for driving said bevel gear assembly.
3. The autonomous artificial muscle fiber coiler system of claim 2, wherein said two parallel motor drivers vary a speed of driving said third bevel gear.
4. The autonomous artificial muscle fiber coiler system of claim 3, wherein said bevel gear assembly having a material selected from the group consisting of a steel, a brass, an aluminum, and

a plastic.

5. The autonomous artificial muscle fiber coiler system of claim 1, wherein a gear ratio of said first bevel gear and said second bevel gear is 1:1 and a gear ratio of said third bevel gear is from 1:1 to 2:1.
6. The autonomous artificial muscle fiber coiler system of claim 5, wherein said first bevel gear and said second bevel gear having from 30 teeth per inch to 40 teeth per inch and said third bevel gear having from 15 teeth per inch to 20 teeth per inch.
7. The autonomous artificial muscle fiber coiler system of claim 6, wherein said monofilament fiber selected from the group consisting of a nylon, a polyester, a Kevlar, a conductive polymer, an elastomer, and a carbon fiber filament.
8. The autonomous artificial muscle fiber coiler system of claim 7, wherein said power supply unit providing electric power to the muscle fiber coiler system.
9. The autonomous artificial muscle fiber coiler system of claim 8, wherein said power supply unit having said plurality of controllers for fine control over a supplied voltage and a current.
10. The autonomous artificial muscle fiber coiler system of claim 6, wherein said first bevel gear and said second bevel gear rotate synchronously.
11. The autonomous artificial muscle fiber coiler system of claim 10, wherein said third bevel gear rotates non-synchronously with said first bevel gear and said second bevel gear.
12. A method of producing an artificial muscle fiber, the method comprising the steps of; providing a motor-driven spool holder, a monofilament fiber, a power supply unit, a plurality of controllers, a pulley system, and a bevel gear assembly, wherein said bevel gear assembly having a first bevel gear and a second bevel gear rotating for a twisting movement of said monofilament fiber, wherein said bevel gear assembly having a third bevel gear disposed between said first bevel gear and said second bevel gear, further wherein said first bevel gear and said second bevel gear are larger than said third bevel gear; holding said monofilament fiber with said motor-driven spool holder; feeding said monofilament fiber to said pulley system of the artificial muscle fiber coiler system; tensioning dynamically said monofilament fiber with said pulley system including a plurality of pulleys; twisting said monofilament fiber; rotating synchronously said plurality of pulleys for regulating said monofilament fiber dynamic tension and for controlling a length of said monofilament fiber for forming an artificial muscle fiber; twisting said monofilament fiber into a coiled structure with said bevel gear assembly; synchronizing rotation of said first bevel gear and said second bevel gear with said third bevel gear; and controlling said twisting of said monofilament fiber with said third bevel gear.
13. The method of producing an artificial muscle fiber of claim 12, wherein a gear ratio of said first bevel gear and said second bevel gear is 1:1 and a gear ratio of said third bevel gear is from 1:1 to 2:1.
14. The method of producing an artificial muscle fiber of claim 13, wherein said first bevel gear and said second bevel gear having from 30 teeth per inch to 40 teeth per inch and said third bevel gear having from 15 teeth per inch to 20 teeth per inch.
15. The method of producing an artificial muscle fiber of claim 14, wherein said monofilament fiber selected from the group consisting of a nylon, a polyester, a Kevlar, a conductive polymer, an elastomer, and a carbon fiber filament.
16. The method of producing an artificial muscle fiber of claim 14, wherein said first bevel gear and said second bevel gear rotate synchronously.
17. The method of producing an artificial muscle fiber of claim 16, wherein said third bevel gear rotates non-synchronously with said first bevel gear and said second bevel gear.
18. A method of producing an artificial muscle fiber, the method comprising the steps of: providing a motor-driven spool holder, a monofilament fiber, a power supply unit, a plurality of controllers, a pulley system, and a bevel gear assembly, wherein said bevel gear assembly having a first bevel gear and a second bevel gear rotating for a twisting movement of said monofilament fiber, wherein

said bevel gear assembly having a third bevel gear disposed between said first bevel gear and said second bevel gear, further wherein said first bevel gear and said second bevel gear are larger than said third bevel gear; holding said monofilament fiber with said motor-driven spool holder; unwinding said monofilament fiber from said motor-driven spool holder; transferring said monofilament fiber to said pulley system; feeding said monofilament fiber to said pulley system of an artificial muscle fiber coiler system; tensioning dynamically said monofilament fiber with said pulley system including a plurality of pulleys; rotating said plurality of pulleys for regulating said monofilament fiber dynamic tension; controlling a length of said monofilament fiber for forming an artificial muscle fiber; twisting said monofilament fiber into a helical structure with said bevel gear assembly; synchronizing rotation of said first bevel gear and said second bevel gear with said third bevel gear; and controlling said twisting of said monofilament fiber with said third bevel gear; wherein a gear ratio of said first bevel gear and said second bevel gear is 1:1 and a gear ratio of said third bevel gear is from 1:1 to 2:1, further wherein said first bevel gear and said second bevel gear having from 30 teeth per inch to 40 teeth per inch and said third bevel gear having from 15 teeth per inch to 20 teeth per inch.

19. The method of producing an artificial muscle fiber of claim 18 further comprising the step of: adjusting dynamically a motor speed of said bevel gear assembly to vary a twist density of said monofilament fiber.

20. The method of producing an artificial muscle fiber of claim 18, wherein said first bevel gear and said second bevel gear rotate synchronously, and further wherein said third bevel gear rotates non-synchronously with said first bevel gear and said second bevel gear.
